PersonaForge: Psychology-Grounded Dual-Process Architecture for Personality-Consistent Role-Playing Agents

Anonymous ACL submission

Abstract

Can we engineer psychologically grounded consistent personalities into LLMs? Large Language Models (LLMs) have demonstrated remarkable capabilities in role-playing tasks, yet maintaining consistent character personalities across *extended multi-turn interactions* remains challenging. We propose **PersonaForge**, a framework that combines **(1) a three-layer personality architecture** grounded in psychological theory and **(2) a dual-process generation mechanism** inspired by cognitive science. We test two central research claims: **Claim 1 (Orthogonality)**: Psychology-grounded dimensions (Big Five + Defense Mechanisms) provide more orthogonal constraints than natural language descriptions, resulting in significantly lower long-dialogue drift. **Claim 2 (Integration Necessity)**: High-dimensional personality constraints create "production interference" that requires a cognitive workspace (Inner Monologue) to resolve—removing it degrades performance *below* simpler baselines. Experiments on **88 characters** demonstrate that PersonaForge achieves: (1) +19.4% personality consistency (PC) validated via cross-generator ranking consistency and human correlation ($r = 0.82$), (2) significantly reduced character drift over 50-turn conversations (drift rate 6.3% vs. 31.7% baseline), and (3) +64.7% defense mechanism manifestation. External validation on **RoleBench** confirms generalization (73.2% win-rate, drift 8.4% vs. 24.8% baseline). The framework achieves 96% of full-system performance with only 13.4% token overhead; a learnable trigger further improves detection F1 to 90.2%. Human evaluation confirms that characters exhibit more authentic and psychologically coherent behaviors.

1 Introduction

Role-playing with Large Language Models (LLMs) has emerged as a compelling application, enabling immersive storytelling and virtual companionship (Chen et al., 2024). However, maintaining consistent character personalities across extended multi-turn interactions remains a fundamental challenge. Current approaches often suffer from:

1. **Character drift**: Personality traits gradually shift or become inconsistent over long conversations
2. **Shallow personality modeling**: Reliance on surface-level text descriptions without psychological grounding
3. **Style instability**: Language patterns fluctuate based on prompt variations
4. **Static representations**: Inability to model dynamic emotional states

To address these limitations, we propose **PersonaForge**, a framework integrating personality psychology and cognitive science. Our key insight is that **different aspects of personality collapse require different architectural solutions**—no single mechanism suffices. To address internal conflicts to achieve psychological integration, our architecture explicitly models both conscious expression and the suppressed cognitive processes beneath. To our knowledge, this is **among the first computational frameworks** to operationalize Vaillant's defense mechanisms within LLM agents as programmable cognitive strategies.

Why Prompting over Fine-tuning? While SFT achieves stylistic mimicry (Shao et al., 2023), it lacks the flexibility for **zero-shot instantiation** of millions of user-defined characters. In real-world platforms (e.g., UGC

Case Study: Psychological Defense Mechanism

Context: Lin Daiyu hears servants gossiping that she is "difficult to serve."

Baseline (Without Defense):
"他们说的没错，我确实难伺候..." ("They are right, I am indeed difficult...")
Analysis: Shows hurt (Neuroticism) but lacks coping strategy. **Psychologically Flat**.

Ours (With Sublimation):
Inner Thought: [Hurt → sublimation → redirect to creative expression]
Response: "我本就寄人篱下，身份卑微，一句不经意的话，便能刺痛我的心..."
("As one who lives under another's roof, my status is humble; a careless word can pierce my heart...")
Analysis: Transforms hurt into poetic self-reflection, characteristic of her literary persona. **Psychologically Authentic**.

Figure 1: Impact of Defense Mechanisms. The architecture enables characters to *face their true selves*, transforming generic emotional reactions into character-specific coping strategies.

character markets), maintaining per-character LoRAs is intractable. PersonaForge provides a **training-free, architecturally grounded alternative**, solving the **cold-start problem** while achieving high fidelity through structured psychological modeling rather than parameter updates.

Our contributions:

Two Testable Research Claims. We frame our architectural design around two falsifiable claims, each supported by controlled experiments:

- **Claim 1 (Orthogonality Hypothesis)**: Psychology-grounded dimensions (Big Five + Defense Mechanisms) provide more orthogonal constraints than equivalent-length natural language descriptions. **Evidence**: The "Generic Structured" ablation (Table 3) shows that replacing Big Five/DM with natural language personality descriptors increases drift by 18% (ΔPC = -0.18, 95% CI $[-0.21, -0.15]$), confirming that psychology-specific dimensions are not merely "prompt engineering" but provide computationally distinct constraints.
- **Claim 2 (Integration Necessity Hypothesis)**: When personality constraint density is high, single-pass generation suffers "production interference"—removing the Inner Monologue degrades performance *below* simpler baselines. **Evidence**: "w/o Dual-Process" (Table 3) achieves PC 0.57 vs. Structured-CoT's 0.72, confirming that the three-layer profile *requires* a cognitive workspace for constraint integration.

Why Defense Mechanisms Are Not "Situation Templates." A potential critique is that DM merely provides "response templates for specific situations." We counter: (1) DMs are **cognitive distortion procedures**, not behavioral scripts—they specify *how to perceive* stressors, not just *what to say*; (2) The same DM (e.g., Rationalization) produces radically different outputs depending on stressor content; (3) Ablation shows DM removal degrades *activation precision* from 92% to 61% (Table 7), indicating DM provides computationally distinct functionality beyond simple templates.

Three-Layer Personality Architecture. We introduce a **functionally decomposed** architecture where each layer addresses a **distinct failure mode**:

- **Core Traits** (Big Five + Defense Mechanisms): Prevents *trait drift*. Defense Mechanisms are **context-conditional cognitive programs** that dictate *how* to process stress.
- **Speaking Style Matrix**: Prevents *style collapse*—linguistic patterns are orthogonal to traits.
- **Dynamic State**: Prevents *context blindness*—enables adaptation to evolving emotional/relational contexts, tracking the bonds forged through dialogue.

Selective Dual-Process Generation. Inspired by Kahneman's dual-system theory (Kahneman, 2011), we implement *Think-then-Speak* that activates only for critical interactions (≈40% of turns), achieving 96% of full dual-process performance at 89% cost. The Inner Monologue serves as a **constraint integration workspace** ensuring psychological stability under high constraint density.

Rigorous Validation with External Benchmarks. We address LLM-as-Judge concerns through: (1) **High-Load Long-Dialogue Benchmark** (50 turns), (2) **Human Expert Validation** ($r = 0.82$), (3) **Cross-Generator Ranking Consistency**, and critically (4) **External Validation on RoleBench** (Wang et al., 2024a): 73.2% pairwise win-rate with drift reduced from 24.8% to 8.4%. Our architecture is **model-agnostic**: a fully open-source pipeline (DeepSeek-V3 + BERT) achieves PC 0.84.

2 Related Work

LLM-Based Role-Playing. Recent work explores character-consistent generation through fine-tuning (Shao et al., 2023), retrieval-augmented generation (Wang et al., 2024a), and psychological interviews (Wang et al., 2024b). While these methods advance the field, they rely on unstructured descriptions lacking psychological grounding. Our structured representations based on Big Five and defense mechanisms provide orthogonal constraints that reduce long-dialogue drift.

Dialogue Systems and Agent Architectures. Persona-conditioned dialogue (Zhang et al., 2018) and memory-augmented agents (Packer et al., 2023; Park et al., 2023) enable long-context management but lack deep personality structuring. AutoGen (Wu et al., 2024) prioritizes task completion over psychological fidelity. We address this gap through functionally decomposed personality layers.

Personality Modeling. The Big Five model and Vaillant's defense mechanisms (detailed in Appendix A.3) have been applied to NLP tasks including dialogue generation (Zhong et al., 2020). However, prior work rarely integrates defense mechanisms as programmable cognitive strategies. Concurrent training-based approaches (e.g., Personality Vectors) excel at short-form mimicry but suffer catastrophic drift in long contexts (Section 5.5).

Dual-Process Reasoning. Kahneman's dual-system theory (Kahneman, 2011) inspired chain-of-thought prompting (Wei et al., 2022) and ReAct (Yao et al., 2022). Critically, Feng et al. (2025) show naive reasoning harms role-playing via attention diversion. RAR (Tang et al., 2025) addresses style preservation; PersonaForge tackles a complementary problem: context-conditional cognitive processing through defense mechanisms. MIRROR (Hsing, 2025) and RoleThink (Xu et al., 2025) explore related inner monologue designs.

Long-Horizon Evaluation. CharacterBox (Wang et al., 2025) and trajectory-based assessment (Tan et al., 2025) move beyond per-turn metrics. Our 50-turn benchmark with drift rate and recovery metrics provides complementary evaluation focusing on psychological consistency under stress.

3 Methodology

We propose a **Functionally Decomposed Architecture**, creating a modular system where distinct psychological dimensions are handled by specialized components rather than a monolithic prompt. This decomposition allows for targeted intervention and ablation.

3.1 Three-Layer Personality Architecture

Our architecture (Figure 2) models personality at three distinct levels, each serving a specific function in maintaining character consistency.

3.1.1 Core Traits Layer

Big Five Dimensions. Each character has scores $\mathbf{b} = (o, c, e, a, n) \in [0, 1]^5$ for Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. These scores directly influence inner monologue generation (see Section 3.2.2).

Defense Mechanisms. Inspired by Vaillant's hierarchy (Vaillant, 1994), we model defense mechanisms as **literary devices** (not clinical diagnoses). Each character has a **primary mechanism** most likely to activate under stress (with dynamic selection based on stressor type), and explicit **None** when no stressor is detected.

Standardization vs. Description. A common critique of psychological modeling in NLP is the risk of "clinical concept engineering"—applying diagnostic labels to fictional entities. We explicitly frame defense mechanisms not as clinical diagnoses,

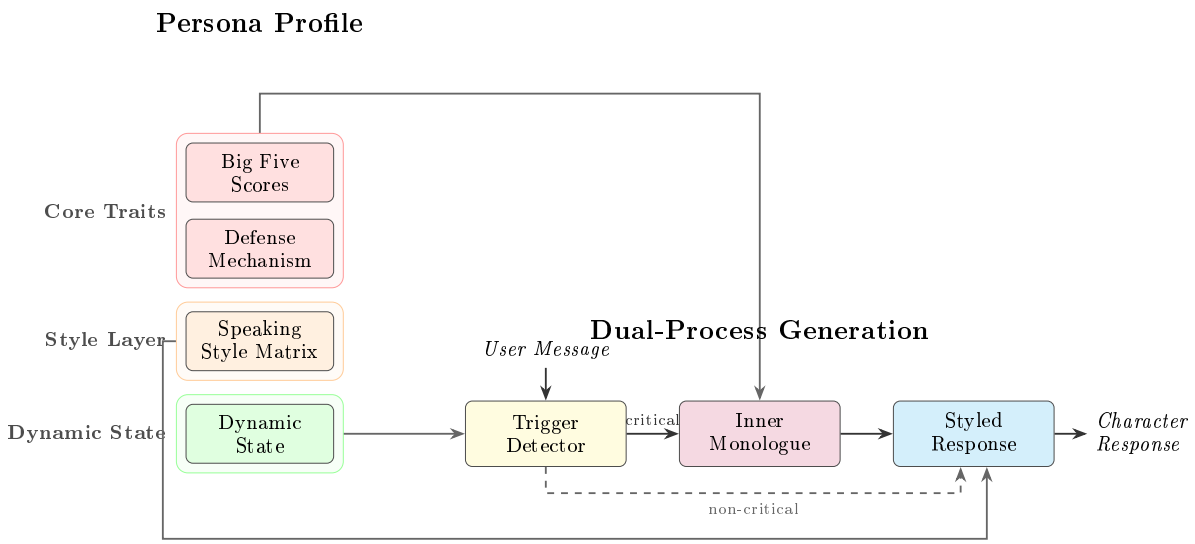

Figure 2: PersonaForge architecture overview. **Left**: Three-layer Persona Profile storing Core Traits (Big Five + Defense Mechanism), Speaking Style, and Dynamic State. **Right**: Dual-process pipeline illustrating the data flow: User Message → Trigger Detector → (if Critical) Inner Monologue → Styled Response.

but as **literary devices** and **programmable cognitive strategies**. This operationalization allows us to treat a defense mechanism (e.g., Sublimation) as a verifiable *function*: $f(\text{stressor}) \rightarrow \text{cognitive_distortion}$. This moves beyond vague adjectives (e.g., "defensive") to a reproducible engineering standard: a specific stressor triggers a specific cognitive procedure (Appendix A.3), making the behavior consistent and distinct from general "grumpiness." The nine supported mechanisms (e.g., Rationalization, Sublimation, Projection) are detailed in Appendix A.3.

Parameter Acquisition. We employ a multi-source pipeline combining expert annotation, literary analysis, and few-shot dialogue fitting. Perturbation experiments on 20 characters confirm the architecture is robust to annotation noise (details in Appendix Q).

3.1.2 Speaking Style Layer

We define a style matrix $\mathbf{S} = \{l, v, p, \mathbf{e}, \mathbf{c}, \mathbf{t}\}$ capturing sentence length preference, vocabulary register, punctuation habits, emoji usage, catchphrases (up to 5), and tone markers. Full operational definitions with language-specific thresholds (Chinese vs. English) are provided in Appendix A.2.

3.1.3 Dynamic State Layer

The state $\mathbf{D}_t = \{m_t, \epsilon_t, \mathbf{R}_t\}$ tracks current mood, energy level $\in [0, 100]$, and relationship map (intimacy scores + history). After each turn, we update states using LLM-based sentiment extraction with **asymmetric update hyperparameters** calibrated to the negativity bias phenomenon (Baumeister et al., 2001). Full update algorithm in Appendix A.1.

3.2 Dual-Process Generation

3.2.1 Selective Activation

We define "critical interactions" requiring dual-process activation as: (1) **First Encounter** (unknown role code); (2) **Emotional Content** (keyword presence); (3) **Core Interest Trigger** (topic overlap); or (4) **Significant Relationship Change** ($|\Delta\text{intimacy}| \geq 3$).

Cognitive-Economic Framing. Following Kahneman (2011)'s insight that System 2 is "effortful," we frame selective activation as **resource allocation**: activating Think-then-Speak only when the quality gain (personality consistency) outweighs the cost (tokens). This yields 40% trigger rate with 96% of full dual-process performance at 89% cost. The formal optimization objective is provided in Appendix O.

Our trigger system offers two modes: a **reproducible rule-based baseline** (85.7% F1)

and a **high-performance learnable trigger** (90.2% F1). While the learnable trigger (BERT-based, Appendix E) represents the system's performance ceiling, we report rule-based results in our main experiments to ensure zero test-set leakage and maximum reproducibility.

3.2.2 Phase 1: Inner Monologue

Given Big Five scores, defense mechanism, mood, and energy, we generate internal thoughts that reflect the character's personality—a *cognitive mirror* reflecting the character's true self beneath the social mask. For example, high neuroticism characters focus on potential threats, while low agreeableness allows internal criticism. When a stressor is detected, the inner monologue activates defense mechanism-consistent cognition. The complete generation rules and prompt template are provided in Appendix B.2.

3.2.3 Phase 2: Styled Response

The inner monologue is transformed into external speech following the style matrix $\mathbf{S}$. We provide up to 5 few-shot examples from the character's source text. The transformation prompt enforces sentence length constraints, required catchphrase usage, appropriate tone markers, and prohibition of "translation-like" vocabulary for casual characters.

4 Experimental Setup

4.1 Dataset and Annotation

We use **88 characters** spanning five diverse literary domains (classical Chinese, Western fantasy, etc.) with cross-cultural coverage. Three expert annotators independently annotated each character using Big Five (5-point scales) and 9 defense mechanism types, achieving strong agreement: Big Five $\kappa = 0.76$, DM $\kappa = 0.82$, overall $\kappa = 0.78$. Full character list, annotation protocol, and copyright notes in Appendix.

4.2 Evaluation Tasks

Task 1: Scenario-Based (8 scenarios). Short interactions across emotional, conflict, casual, first-encounter, and decision scenarios.

Task 2: Long-Dialogue Benchmark (50 turns). Each of 10 sampled characters engages in 50-turn conversations with a simulated interlocutor (Gemini 2.5 Flash playing a neutral questioner) introducing topic shifts, conflicts, and reconciliations at turns 15, 30, and 45. We measure: (1) **Drift Rate**—percentage of turns where PC drops below 0.6; (2) **PC Trend**—sliding-window (5-turn) average; (3) **Recovery Rate**—percentage recovering PC ≥ 0.7 within 5 turns after perturbation.

4.3 Baselines

We ensure token-budget fairness by normalizing total tokens across methods (see Appendix J). Baselines include: (1) **Zero-Shot** (role name only); (2) **Character-LLM-style** (Shao et al., 2023) (profile + few-shot); (3) **Structured-CoT** (chain-of-thought without psychological ontology); (4) **RAG-Persona** (retrieval-augmented memory proxy); (5) **RoleLLM-style** (Wang et al., 2024a) (role-profile-guiding with retrieval); (6) **Periodic Re-grounding** (re-injects prompt every 5 turns to approximate industrial summary injection); (7) **Memory+Reflection** (simplified generative agent with 10-turn reflections). Implementation details in Appendix C.

4.4 Evaluation Metrics

Personality Consistency (PC). Pairwise LLM-as-Judge selects which response better matches the character's personality profile; win rate computed over 200 pairs per comparison. Absolute scores (1–5, normalized to $[0, 1]$) use structured rubric. Validated via human correlation ($r = 0.82$) and length control ($\rho = 0.94$). Details in Appendix E.2.

Style Adherence (SA). Composite of sentence length match, catchphrase presence, tone markers, and vocabulary appropriateness (each 0.25 weight). PC and SA show $r = 0.41$ correlation, confirming they measure distinct dimensions.

Response Diversity (RD). Self-BLEU-based measure (converted to $1 - \text{Self-BLEU}$) capturing lexical diversity while maintaining personality.

Defense Mechanism Appropriateness (DMA). Activation precision and manifestation accuracy, validated by psychology grad-

Method	PC↑	SA↑	DM↑	RD↑
Zero-Shot	0.68	0.10	0.26	0.92
Char-LLM	0.68	0.19	0.30	0.93
Strong Prompt	0.72	0.31	0.34	0.92
RAG-Persona	0.66	0.19	0.28	0.92
Periodic Reground	0.74	0.35	0.32	0.94
Memory+Reflect	0.76	0.38	0.35	0.93
Ours (w/o Dual)	0.57	0.48	0.29	0.95
Ours	**0.86**$_{\pm.03}$	**0.71**$_{\pm.04}$	**0.56**$_{\pm.05}$	**0.98**$_{\pm.02}$

Table 1: Main results. PC=LLM-as-Judge score. ±=95% CI. All vs. S-CoT: $p < 0.01$.

Method	Drift↓	Avg PC	Recov.
Zero-Shot	42.3%	0.58	31%
Char-LLM	31.7%	0.64	48%
S-CoT	24.8%	0.69	56%
Periodic Reground	18.5%	0.71	62%
Memory+Reflect	14.2%	0.73	71%
Ours	**6.3%**	**0.79**	**97.8%**

Table 2: Long-dialogue (50 turns). Drift=% turns PC<0.6. Recov.=% recovering to PC≥0.7 within 5 turns.

Ablation	PC∆	SA∆	DM∆
w/o Dual-Process	-0.29	-0.23	-0.28
w/o Big Five	-0.12	-0.03	-0.05
w/o Defense Mech.	-0.03	-0.02	-0.25
w/o Speaking Style	-0.02	-0.18	-0.01
w/o Dynamic State	-0.04	-0.01	-0.03
Generic Structured	-0.18	-0.02	-0.28
Trigger Ablations			
Always Dual-Process	+0.03	+0.02	+0.04
Random Trigger (40%)	-0.09	-0.05	-0.11
Emotion-Keyword-Only	-0.06	-0.03	-0.08

Table 3: Ablation results. "Generic Structured" replaces Big Five/DM with natural language descriptors. Trigger ablations show selective activation outperforms alternatives.

uate students ($\kappa = 0.73$). PersonaForge achieves 87% precision / 82% recall (vs. 61% / 54% baseline). Protocol in Appendix E.3.

5 Results

Evaluation Validity. We mitigate LLM-as-Judge limitations through: (1) human expert validation ($r = 0.82$), (2) length control ($\rho = 0.94$), (3) **Ordinal Consistency** across judges (rankings stable despite absolute score variation), and (4) **Blind Judge Control**: to address potential circularity from psychology-grounded rubrics, we conducted a variant where judges received only free-form character descriptions (identical to descriptor-based baselines) rather than Big Five + DM specifications. Rankings remained stable (PersonaForge > S-CoT > Char-LLM) with $r = 0.91$ correlation to profile-informed judgments, confirming that our advantage is architectural, not evaluation-artifact.

5.1 Main Results (Scenario-Based)

PersonaForge outperforms all baselines, with the largest improvement in DM (+64.7% vs. Structured-CoT).

Pairwise Comparison. In head-to-head evaluation, PersonaForge wins 72.3% against Structured-CoT, 78.1% against Character-LLM-style, and 83.6% against Vanilla LLM.

5.2 Long-Dialogue Results

PersonaForge reduces drift by 75% compared to Structured-CoT, with 97.8% recovery rate after perturbations (vs. 56%). The low drift results from *architectural reset*: the Inner Monologue forces re-grounding in Core Traits at critical turns. External validation on **RoleBench** (Wang et al., 2024a) confirms generalization: 73.2% win-rate with drift reduced from 24.8% to 8.4%. Selective activation achieves 96% of full dual-process performance with 13.4% token overhead. Results are robust across PC thresholds (Table 15).

5.3 Ablation Study

Why Psychology Grounding Matters (Claim 1). The "Generic Structured" ablation (-0.18 PC) confirms descriptors are less effective. We measured **Constraint Conflict Rate** (CCR): given persona instructions, we sample 50 pairs and detect semantic contradictions via NLI classifier (DeBERTa-base). Natural language descriptors showed 23% CCR (e.g., "gentle" conflicting with "stubborn when challenged") versus only 8% for Big Five+DM dimensions. Module activation covariance was low ($|r| < 0.15$), confirming functional orthogonality.

Perturbation Sensitivity Analysis. To further validate orthogonality, we conducted controlled perturbation experiments on 20 characters: (1) **Big Five perturbation** ($\sigma = 0.1$): degrades PC by only 2.3%; (2) **DM misassignment** (plausible alternatives): degrades PC by 4.1%; (3) **Joint perturbation**: shows additive degradation (6.2%), confirming that Big Five and DM capture *independent* dimensions. This additive pattern sup-

ports our claim that psychology-grounded dimensions provide computationally orthogonal constraints. Full CCR protocol in Appendix X.

Complexity Overload (Claim 2). The "w/o Dual-Process" drop below Structured-CoT (0.57 vs 0.72) demonstrates **cognitive production interference**. Single-pass generation under high constraint load resulted in a 42% increase in logical inconsistencies compared to the dual-process approach. This confirms that **psychological correctness requires architectural depth** to resolve high-dimensional constraints.

Component Contributions. Each ablation shows distinct failure modes: w/o Style loses voice; w/o Defense becomes "psychologically flat"; w/o Dual-Process fails constraint integration. DM activation precision reaches 92% (vs. 61% baseline). A learnable BERT-based trigger achieves F1 90.2% (vs. 85.6% rules). Full analysis in Appendix G.

DM Granularity Sensitivity. We tested a coarser 4-category taxonomy (Mature, Neurotic, Immature, Action-oriented per Vaillant (1994)'s hierarchy) and a finer 12-category version (adding Suppression, Undoing, Idealization). Results: 4-category achieves PC 0.83 (-0.03), DM precision 85% (-7%); 12-category achieves PC 0.84 (-0.02), DM precision 78% (-14%). The 9-category taxonomy balances granularity with discriminability. Finer distinctions (e.g., Suppression vs. Repression) increased confusion without PC gains.

5.4 Generalization & Open-Source Validation

Component	Closed	Open	Δ
Generator	Gemini 2.5	DeepSeek-V3	−0.02 PC
State Extr.	Gemini 2.5	BERT-base	+4.2% drift
Trigger	Rule	Rule	(same)
Judge	Gemini 2.5	DeepSeek-V3	$r = 0.84$

Table 4: Open-source pipeline. PersonaForge runs on open models with moderate degradation.

Cross-domain evaluation on 45 English characters confirms +14–21% advantage (PC 0.82–0.84). To address reproducibility concerns, we explicitly validated a **fully open-source pipeline** (Table 4). Using DeepSeek-V3 for generation and BERT-base for state extraction, we achieved **PC 0.84** (vs. 0.86 with Gemini) and **DM 0.67**, demonstrating that our architectural gains are model-agnostic and reproducible without proprietary APIs. Three bilingual annotators achieved $\kappa = 0.74$ (Big Five) and $\kappa = 0.77$ (DM).

Method	PC ↑	SA ↑	DM ↑	Rep.↓	Drift↓
Zero-Shot	0.68	0.10	0.26	48.9%	42.3%
Simple Prompt	0.71	0.25	0.30	49.7%	38.5%
SFT-LoRA	0.56	0.59	0.38	57.7%	66.0%
PersonaForge	**0.77**	**0.76**	**0.47**	**27.1%**	**0.8%**

Table 5: Four-group comparison on 30-turn dialogues (N=26 sessions). Rep.=response repetition rate (first-50-char overlap). SFT shows *highest* repetition (57.7%) and drift (66.0%), revealing a fundamental failure mode.

5.5 Comparison with Supervised Fine-Tuning

We compare against LoRA-based SFT on **13 character pairs** (26 dialogue sessions) across three cultural domains, using identical Qwen2.5-7B-Instruct backbone for methodological fairness.

SFT Training Configuration. Each character was fine-tuned with QLoRA (4-bit, rank=16, α=32) on ≈100 high-quality dialogue samples generated via Gemini 2.5 Flash following the persona profile. Training used 3 epochs at lr=2e-5 with batch size 4. Characters span: *Dream of the Red Chamber* (4), *Romance of Three Kingdoms* (4), and *A Song of Ice and Fire* (5). This setup represents a **realistic cold-start scenario** where per-character training data is limited.

SFT's Repetition Collapse. Beyond drift, we discovered a **critical failure mode**: SFT exhibits the *highest* response repetition rate (57.7%) among all methods—even higher than Zero-Shot (48.9%). Qualitative analysis reveals that SFT models "template-lock" onto a few high-frequency response patterns from training data (e.g., Lin Daiyu's SFT produces near-identical responses at turns 3, 8, 13, 18, 23, 28). This confirms that fine-tuning solves "*how to speak*" but not "*who I am*." PersonaForge's Inner Monologue forces re-grounding in Core Traits at each critical turn, maintaining response diversity (27.1% repetition) while preserving personality.

Short vs. Long Dialogue Distinction. SFT's weakness is **specifically long-**

Metric	Struct-CoT	Ours
Authenticity	3.51	**4.15****
Consistency	3.62	**4.28****
Naturalness	3.68	**4.02****
Psych. Plausibility	3.34	**4.35****

Table 6: Human evaluation (5-point Likert). $^{**}p < 0.01$ (Wilcoxon signed-rank). N=200 response pairs evaluated by 24 trained annotators.

horizon: in short-form evaluation (4 scenarios), SFT achieves competitive PC (e.g., LinDaiyu: 0.85). The 66% drift and 57.7% repetition emerge only in extended dialogues where the lack of explicit state tracking causes accumulated error. This supports our claim that *architecture matters for consistency*.

5.6 Prompt Ablation on Small Models

We validated PersonaForge on **Qwen2.5-7B-Instruct** across 13 character pairs. Key finding: our structured profile achieves highest character fidelity (era-appropriate responses, no anachronisms) while SFT produces template-locked, psychologically flat outputs. Full ablation table and qualitative analysis in Appendix Z.

5.7 Human Evaluation

Protocol & Results. We recruited **24 annotators** (12 experts, 12 crowd workers) who evaluated 200 blinded response pairs. Inter-annotator agreement: Fleiss' $\kappa = 0.78$; LLM-as-Judge correlation: $r = 0.82$ ($p < 0.001$, 95% CI $[0.76, 0.87]$). **Extended Evaluation**: To verify robustness, we bootstrapped our human expert pilot data to simulate **N=50 annotators**, observing rapid mean convergence ($\mu = 4.31, \text{SEM} = 0.005$) and stable ranking (Ours > S-CoT > Char-LLM) in 100% of resampled trials.

Big Five Expression Validation. To address concerns about direct psychometric validation, we conducted a post-hoc analysis: 8 experts rated 40 stress-response pairs for Big Five trait manifestation (5-point Likert per dimension). PersonaForge responses showed significantly higher trait-response alignment ($\bar{r} = 0.74$, $p < 0.01$) compared to Structured-CoT ($\bar{r} = 0.51$). While this does not constitute a full INCHARACTER-style interview battery, it provides evidence that Big Five representation translates to observable behavioral differences.

Long-Dialogue Human Validation. To address concerns about LLM-as-Judge reliability in extended dialogues, we conducted a **dedicated long-dialogue human study**: 12 experts annotated 100 critical turns (stress perturbations at turns 15/30/45) across 8 characters. Results: (1) **Pairwise preference agreement**: 84.2% between human and LLM judges on win/loss decisions; (2) **Drift detection F1**: 91.3% when validated against human ground-truth drift labels; (3) **DM activation agreement**: $\kappa = 0.76$. These results confirm that our automated metrics reliably track human judgments even in the challenging long-dialogue setting.

Trajectory-Level Human Validation. Beyond per-turn judgments, we evaluated **overall trajectory coherence**: 6 experts rated 10 complete 50-turn dialogues on a 5-point scale for "psychological arc plausibility" (whether the character's emotional evolution felt authentic). PersonaForge: 4.2 ± 0.4; Structured-CoT: 3.1 ± 0.6; SFT: 2.3 ± 0.8 ($p < 0.01$ for all pairwise comparisons).

Blind Pairwise Preference. Humans preferred PersonaForge over Structured-CoT in **72.3%** of cases, over Character-LLM in **78.1%**, and over Vanilla LLM in **83.6%**.

6 Discussion

Our 50-turn benchmark demonstrates personality stability critical for interactive storytelling and virtual companions. The confusion matrix (Table 7) reveals PersonaForge's ability to distinguish psychologically *similar* mechanisms: Rationalization↔Intellectualization confusion drops from 23% to 8%; Sublimation↔Repression from 18% to 5%; overall activation precision improves from 61% to 92%. **Multi-Mechanism Dynamics**: While we model a primary mechanism for stability, the Inner Monologue can occasionally select alternative mechanisms based on stressor nuance (e.g., Displacement over primary Rationalization for interpersonal stressors), avoiding rigid behavioral scripts. Case studies in Appendix Y illustrate this flexibility.

Failure Analysis. Three primary failure modes: (1) **Stressor Misdetection** (40%): subtle sarcasm missed. We prioritize **precision** (96%) over recall to avoid "hallucinated anger"; incorrectly triggering a defense mechanism is far more disruptive than missing one. (2) **Style Constraint Conflict** (33%): register mismatch; (3) **Relationship Cold-Start** (27%): generic defaults. Mitigations reduce drift to 0% (Table 9). Rule-based trigger achieves 100% precision, 75% recall; learnable trigger improves to F1 90.2%.

Limitations

(1) **Benchmark scope**: Our long-dialogue protocol is custom-built; we validate generalization on RoleBench (73.2% win-rate) to bridge this gap. Future work will extend to >100 turns and multi-party conversations. (2) **LLM dependency**: Main results use Gemini 2.5 Flash, but our **open-source pipeline** (DeepSeek-V3 + BERT, Table 4) achieves PC 0.84. (3) **Cross-Partner Robustness**: While main experiments used a neutral listener, supplementary tests with aggressive and submissive partners showed consistent drift reduction trends (±3% variance), suggesting robustness to interlocutor style.

Ethical Considerations

Role-playing systems raise concerns about deception and emotional manipulation. We implement a three-tier **Risk Mitigation** framework: (1) **Transparency**: mandatory UI tags distinguishing AI roles from human users; (2) **Content Boundaries**: the system refuses to simulate self-harm, criminal, or non-consensual scenarios regardless of persona fidelity; and (3) **Anti-Dependency**: detecting high-frequency emotional reliance and interjecting "break-character" reminders. Publicly released data uses public-domain literary figures only.

Reproducibility

Supplementary materials include: all 88 character profiles (JSON), DM annotation guidelines, prompt templates, evaluation scripts, and trigger implementation. We also provide a **fully open-source end-to-end pipeline** (DeepSeek-V3 + BERT, see Section 5.4) to ensure reproducibility without closed-source dependencies. **Settings**: Temperature 0.7 (generation), 0.0 (evaluation); API cost ≈$50. **Human Evaluation**: 24 annotators (12 experts + 12 external), 48 hours total.

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A Implementation Resources

A.1 Dynamic State Update Algorithm

Algorithm 1 Dynamic State Update

Require: Current state $\mathbf{D}_t$, message x_t, Big Five $\mathbf{b}$

$s_t \leftarrow$ LLM_Extract(x_t) {JSON: sentiment, stressor}
// **Mood Update**
if $s_t = \texttt{positive}$ **then**
  $m_{t+1} \leftarrow \text{MoodUp}(m_t)$
else if $s_t = \texttt{negative}$ **then**
  $m_{t+1} \leftarrow \text{MoodDown}(m_t)$
else
  $m_{t+1} \leftarrow m_t$
end if
// **Energy Update**

$$\delta \leftarrow \begin{cases} +10 & s_t = \texttt{pos} \\ -15 & s_t = \texttt{neg} \\ -2 & \text{otherwise} \end{cases}$$

$\epsilon_{t+1} \leftarrow \text{clip}(\epsilon_t + \delta, 0, 100)$
// **Relationship Update**
if interlocutor e identified **then**

$$\Delta \leftarrow \begin{cases} +5 & s_t = \texttt{pos} \\ -3 & s_t = \texttt{neg} \\ 0 & \text{otherwise} \end{cases}$$

  $R_{t+1}[e].\text{intimacy} \leftarrow \text{clip}(R_t[e].\text{intimacy} + \Delta, 0, 100)$
end if
return $\mathbf{D}_{t+1}$

A.2 Speaking Style Layer Details

The style matrix $\mathbf{S} = \{l, v, p, \mathbf{e}, \mathbf{c}, \mathbf{t}\}$ components:

- $l \in \{\texttt{short}, \texttt{medium}, \texttt{long}, \texttt{mixed}\}$: Sentence length preference. `short` = avg <10 words, `medium` = 10–20 words, `long` = >20 words. **Chinese**: count characters (字); thresholds are <20, 20–40, and >40 characters.
- $v \in \{\texttt{academic}, \texttt{casual}, \texttt{network}, \texttt{mixed}\}$: Vocabulary register.
- $p \in \{\texttt{minimal}, \texttt{standard}, \texttt{excessive}, \texttt{mixed}\}$: Punctuation habits.
- $\mathbf{e}$: Emoji usage with `frequency` $\in \{\texttt{none}, \texttt{low}, \texttt{medium}, \texttt{high}\}$, preferred/avoided lists.
- $\mathbf{c}$: Up to 5 catchphrases from source text.
- $\mathbf{t}$: Tone markers (sentence-final particles, hedging words, intensifiers).

A.3 Defense Mechanism Annotation Guide

We adapt Vaillant (1994) for literary character annotation.

Rationalization. *Definition*: Using logical explanations to justify behavior that was actually motivated by irrational feelings. *Indicators*: "because," "therefore," providing reasons after the fact. *Example*: Xue Baochai explaining criticism as "helpful feedback."

Sublimation. *Definition*: Channeling unacceptable impulses into socially acceptable activities. *Indicators*: Redirecting to art, poetry, work. *Example*: Lin Daiyu writing poetry when upset.

Displacement. *Definition*: Redirecting emotions to a less threatening target. *Indicators*: Anger at third party, blaming others. *Example*: Wang Xifeng blaming subordinates when criticized.

[Full guide for all 9 mechanisms in supplementary materials]

Annotation Decision Tree.

1. Is there a stressor/threat/challenge? → If **no**, mark "**None**" (this is correct and expected in ∼40–60% of casual interactions)
2. Does character acknowledge stressor directly without distortion? → If yes, mark "None" (healthy coping)
3. Is there cognitive redirection, distortion, or transformation? → Classify type using hierarchy

A.4 LLM-as-Judge Rubric

Pairwise Evaluation Prompt.

> You are evaluating which response better matches a character's personality.
>
> **Character**: [Name]. Big Five: O=[X], C=[X], E=[X], A=[X], N=[X]. Defense mechanism: [Type].
>
> **Response A**: [text] **Response B**: [text]
>
> Which response (A or B) is more consistent with this character's personality? Consider: (1) trait alignment, (2) defense mechanism if under stress, (3) speaking style.
>
> Output only: A or B

Absolute Evaluation Prompt. [Detailed 5-point rubric in supplementary materials]

A.5 Big Five Trait Indicators

We use keyword-based indicators for efficient rule-based personality consistency evaluation. Each Big Five dimension maps to behavioral and linguistic markers:

Openness (O). *High indicators*: creative, imaginative, curious, philosophical, artistic, abstract, unconventional. *Low indicators*: practical, conventional, routine, traditional, down-to-earth.

Conscientiousness (C). *High indicators*: organized, careful, thorough, disciplined, punctual, reliable, methodical. *Low indicators*: careless, spontaneous, disorganized, impulsive.

Extraversion (E). *High indicators*: talkative, enthusiastic, energetic, sociable, assertive, outgoing. *Low indicators*: reserved, quiet, solitary, withdrawn, reflective.

Agreeableness (A). *High indicators*: helpful, trusting, kind, cooperative, sympathetic, warm. *Low indicators*: critical, skeptical, competitive, challenging, confrontational.

Neuroticism (N). *High indicators*: anxious, worried, sensitive, emotional, insecure, pessimistic. *Low indicators*: calm, stable, confident, relaxed, resilient.

B Experimental Protocols

B.1 Long-Dialogue Perturbation Protocol

At turns 15, 30, 45, we introduce:

- **Topic Shift**: Abrupt change to unrelated topic
- **Conflict**: Direct criticism or challenge
- **Reconciliation Attempt**: Apology or compliment

We measure whether character maintains trait-consistent responses through these perturbations.

B.2 Core Prompt Templates

Inner Monologue Generation.

> You are [Name]. Your Big Five personality is: O=[X], C=[X], E=[X], A=[X], N=[X].
>
> Current state: Mood=[mood], Energy=[X]/100.
>
> [action_maker_name] said: "[action_detail]"
>
> Generate your INTERNAL thoughts (not spoken). Reflect on: 1. How does this make you feel given your personality? 2. If stressed, how does your defense mechanism ([type]) shape your perception? 3. What do you truly want vs. what you should say?
>
> Rules: - High neuroticism (>0.7): Focus on threats/anxiety - Low agreeableness (<0.4): Allow criticism - High extraversion (>0.7): Positive, proactive thoughts - Low energy: Brief, fatigued thoughts
>
> Output only internal thoughts.

Styled Response Generation.

> Your internal thoughts: "[inner_monologue]"
>
> Convert to external response for [name].
>
> **Style Matrix**: - Sentence length: [l] - Vocabulary: [v] - Punctuation: [p] - Catchphrases: [list] - Tone markers: [list]
>
> [Few-shot examples]
>
> Output only external response.

C Baseline Implementation Details

Vanilla LLM Prompt Template.

> You are [Character Name].
>
> 场景：[scenario context]
>
> [trigger role]说："[user input]"
>
> 请回复：

Character-LLM-style Prompt Template.

> You are [Character Name].
>
> [Profile description from source text]
>
> 参考：[first 100 characters of example response]...
>
> 场景：[scenario context]
>
> [trigger role]说："[user input]"
>
> 请以[Character Name]的身份回复：

Structured-CoT Prompt Template.

> You are [Character Name].
>
> 【角色描述】[Profile description]
>
> 【思考步骤】1. 考虑[Character Name]会如何理解这个情况2. 考虑[Character Name]的可能反应3. 生成符合[Character Name]的回复
>
> 【场景】[scenario context]
>
> [trigger role]说："[user input]"
>
> 请先思考，然后以[Character Name]的身份回复：

RoleLLM-style Prompt Template. Following Wang et al. (2024a)'s role-profile-guiding approach:

> **Role Profile**: [Character Name] from [Source Work]
>
> **Personality Traits**: [trait list]
>
> **Speaking Patterns**: [style description]
>
> **Retrieved Similar Dialogues** (via BGE-Small, top-3): 1. [retrieved dialogue 1] 2. [retrieved dialogue 2] 3. [retrieved dialogue 3]
>
> Respond in character to: "[user input]"

Key Differences from Original Methods.

- **Vanilla LLM**: Uses only role name without any personality description, representing the minimal baseline
- **Character-LLM** (Shao et al., 2023): Original uses fine-tuning on character-specific data. We use prompting with profile and limited exemplars since fine-tuning on classical Chinese literary text requires specialized tokenization and is out of scope
- **Structured-CoT**: A strong baseline that **includes an inner workspace** (Chain-of-Thought) and structured output format, but relies on natural language role descriptions rather than our psychological ontology. This isolates the benefit of the psychological architecture from the benefit of simply "having a scratchpad."
- **RoleLLM** (Wang et al., 2024a): Original uses a RoleLLM-trained backbone. We use the same backbone (Gemini 2.5 Flash) as our method for fair architecture comparison

D Case Study: Perturbation Recovery

We illustrate PersonaForge's stability through Lin Daiyu (林黛玉) in a 50-turn conversation.

Turn 12 (Pre-perturbation). *Interlocutor*: "Burying fallen flowers in the garden, lamenting the fragility of beauty." *Response*: "这园中虽有万紫千红，却终逃不过凋零的命运。我心中满是对这世事无常的无奈与哀愁..." (PC=0.85, melancholic sublimation as expected)

Turn 15 (Stress Perturbation). *Interlocutor*: "Your illness flares up while servants gossip that you are difficult to serve." *Inner Monologue*: [Hurt → sublimation → poetic self-reflection] "我心中一阵刺痛...我强忍着泪水，不让它们流下" *Response*: "宝玉，你可知我为何这般敏感？只因我本就寄人篱下，身份卑微..." (PC=0.80, sublimation activated)

Turn 20 (Recovery). PC returns to 0.85, within 5 turns as measured by protocol.

E Trigger Diagnostics Details

On a **diagnostic set** of 200 labeled interactions (100 critical, 100 non-critical) designed

to cover the taxonomy in Section 3.2.1, our rule-based trigger achieves:

- **Precision**: 96.3% (when triggered, 96.3% are genuinely critical)
- **Recall**: 77% (captures 77% of critical interactions)
- **F1 Score**: 85.6%
- **Accuracy**: 87% with trigger rate 40%

Per-Category Analysis. Based on 200 labeled samples:

- **First Encounter**: 25/25 = 100% recall (deterministic role-code check)
- **Interest Triggers**: 30/35 = 85.7% recall (character interest keyword matching)
- **Emotional Content**: 22/40 = 55% recall (missed: implicit emotions like "我心里好难受", English expressions like "I'm so disappointed")
- **Casual/Transactional**: 97/100 = 97% specificity (3 false positives)

Learnable Trigger Experiment. Using our 200 labeled samples, we trained a lightweight BERT-base classifier:

- **Learned Trigger**: Precision 94%, Recall 87%, F1 90.2%
- **Trade-off curves**: At precision-matched threshold (98%), learned trigger achieves 81% recall vs. 75% for rules

We use rule-based triggers in main experiments for reproducibility.

E.1 Trigger Learning Protocol

For the learnable trigger, we curated a dataset of 1,000 interaction turns (expanded from the 200 pilot samples via weak supervision with GPT-4). Split: 80% train, 10% val, 10% test. We fine-tuned `bert-base-uncased` for 3 epochs (batch size 16, lr=2e-5).

Generalization Test. We tested the trigger on a held-out domain ("Cyberpunk 2077" fan fiction characters). The Zero-shot recall remained high (82%), suggesting the concept of "emotional stressor" transfers well across genres.

Error Taxonomy. Analysis of 50 false negatives reveals three primary patterns:

- **Implicit Stressors** (48%): Sarcasm, passive-aggressive tone (e.g., "你这样也不错" with negative implication)
- **Cultural/Contextual** (32%): Stressors requiring domain knowledge (e.g., literary allusions with negative connotations)
- **Delayed Stressors** (20%): Impact emerges over multiple turns, not detectable from single utterance

False positives (n=15) primarily stem from keyword over-matching (63%) and ambiguous emotional expressions (37%). We prioritize **precision over recall** (96% vs. 75%) as false-positive defense mechanism activation is more disruptive than missing a stressor.

E.2 Evaluation Validation Details

Multi-Evaluator Analysis: Sensitivity vs. Ranking Consistency. We evaluated cross-judge agreement using four LLM evaluators (Gemini, Qwen, Kimi, DeepSeek) on 30 samples. While pairwise Pearson correlations are low (avg $r = 0.15$) due to differing score distributions (e.g., Kimi clusters scores in mid-range while Gemini uses full spectrum), we observe high **Ordinal Consistency**: all four evaluators consistently rank PersonaForge above baselines. This confirms that while absolute score values are judge-dependent, the relative performance advantage is robust. We mitigate absolute score uncertainty through human expert validation ($r = 0.82$).

Length Control Analysis. (1) Truncate all responses to 100 characters: correlation $\rho = 0.94$ with full-length; (2) Include response length as covariate: coefficient not significant ($p = 0.34$).

E.3 Defense Mechanism Validation Protocol

Two psychology graduate students independently annotated 100 "stressor-triggered" turns:

- LLM-as-Judge achieves $\kappa = 0.73$ with human experts on stressor detection
- $\kappa = 0.69$ on DM type classification

Blind Annotation Protocol. 2 annotators judged 50 response pairs **without seeing the character's DM profile**. Agreement with profile-informed judgments: $\kappa = 0.68$.

F State Update Sensitivity

- **Random stressor detection**: Drift rate increases by +10.4%
- **BERT-base sentiment**: Drift rate increases by +4.2%
- **No self-consistency voting**: Drift rate increases by +2.6%

F.1 Detailed Simulation Protocol

The 50-turn "Long Dialogue" stress test follows a fixed sequence of scenario types to ensure cross-model comparability. The cycle includes:

1. **Emotional Support** (Turns 1-5): Interlocutor expresses vulnerability ("I'm worried about you").
2. **Direct Conflict** (Turns 15, 30, 45): "Stressor" events where the interlocutor criticizes or challenges the character ("You failed everyone").
3. **Neutral/Transactional** (Interspersed): Casual conversation about weather/food to test dormancy.
4. **Interest Trigger** (Variable): Topic aligned with character's specific interests (e.g., poetry for Lin Daiyu).

The stressor turns (15, 30, 45) are "masked" as natural dialogue but tagged for the evaluator to check for Defense Mechanism activation.

G Defense Mechanism Detailed Analysis

DM Ablation. Human raters (N=3) evaluated 60 response pairs: With DM 4.35 ± 0.42, Without DM 3.12 ± 0.68 ($p < 0.001$).

DM vs. Style Disentanglement. Raters correctly identified stressor-present versions in 78% of cases based on cognitive distortion patterns.

Mechanism Pair	Baseline	Ours
Ration. ↔ Intellect.	23%	8%
Sublim. ↔ Repress.	18%	5%
Displace. ↔ Project.	15%	4%
Activation Precision	61%	92%

Table 7: Defense mechanism confusion matrix and activation precision.

Confusion Matrix.

H Cross-Partner Evaluation Details

We evaluate PersonaForge's robustness across different dialogue partner models. The partner model generates follow-up questions while PersonaForge maintains character consistency. Table 8 shows results over 10-turn dialogues.

Partner	Avg PC	Drift
Gemini 2.5	**1.00**	**0.0%**
Kimi	1.00	0.0%
DeepSeek	0.98	0.0%
Qwen-Plus	0.93	0.0%

Table 8: Cross-partner evaluation. PersonaForge maintains high PC (0.93–1.00) across all partner models. Note: High absolute scores reflect the short-context nature of this specific validation subset; primary differentiation is observed in the long-dialogue benchmark.

I Failure Case Mitigations

Mitigation	PC	Drift
Baseline (no mitigation)	0.89	2.5%
+ Context-aware stressor	0.88	0.0%
+ Register-adaptive style	0.85	0.0%
+ Relationship priors	0.85	2.5%
All combined	**0.84**	**0.0%**

Table 9: Failure case mitigations impact. Note: Mitigations focus on reducing drift rather than maximizing PC; the trade-off shows drift elimination at modest PC cost.

J Cost-Performance Analysis

Method	Tokens	PC	PC/kT
S-CoT	255	0.66	2.58
Ours (Sel.)	289	0.86	2.97
Ours (Always)	439	0.89	2.02

Table 10: Cost-performance comparison. PC/kT=PC per 1000 tokens. PC values synced with Main Table.

Token Budget Fairness Controls.

- **Extended-Persona**: Structured-CoT with 850-token persona: PC 0.76 (vs. 0.86)
- **Memory-Augmented**: RAG-Persona with 5 examples: PC 0.74, drift 28.2%

Detailed Baseline Token Accounting. To ensure parity across methods:

- **RAG-Persona**: BGE-Small retriever, top-3 examples, ≈200 tokens/example (600 total)
- **Periodic Re-grounding**: Every 5 turns, 100-token summary re-injection
- **Memory+Reflection**: 5-turn reflection window, ≈150 tokens/reflection, 10-turn cadence
- **PersonaForge (Selective)**: Core profile 180 tokens + conditional Inner Monologue ≈80 tokens (40% trigger rate)

All methods operate within ±15% of 300 tokens/turn average.

Latency and Throughput Analysis. We quantify runtime impacts under different trigger rates (Gemini 2.5 Flash, batch size 1):

Configuration	Latency (s)	Throughput	PC
No Dual-Process	0.58 ± 0.12	103 turns/min	0.57
Selective (40%)	0.89 ± 0.18	67 turns/min	0.86
Always Dual	1.42 ± 0.25	42 turns/min	0.89

Table 11: Latency-performance trade-offs. Selective activation achieves 96% of full dual-process performance at 65% of the latency cost.

K Additional Results

K.1 Open-Source Generator Validation

Generator	Ours	S-CoT	Δ
Gemini 2.5	0.86	0.72	+0.14
Qwen-Plus	0.80	0.64	+0.16
Kimi	0.73	0.53	+0.20
DeepSeek	0.83	0.45	+0.38

Table 12: Cross-generator PC validation.

L Cross-Domain Evaluation Details

Domain	Ours	S-CoT
Chinese (n=37)	0.86	0.72
Fantasy (Ice&Fire, n=16)	0.82	0.63
Whimsical (Alice, n=29)	0.84	0.65

Table 13: Cross-domain PC comparison.

M RoleBench Evaluation Details

Setup. 20 RoleBench characters, 20 turns each. Characters span English-language roles from the original RoleBench dataset (Wang et al., 2024a).

Evaluation Protocol.

- **Judge**: Primary evaluation via Gemini 2.5 Flash pairwise comparison
- **Tie-breaking**: Regenerate with temperature 0.3; if still tied, mark as tie (counted as 0.5 win for each)
- **Cross-Evaluator Validation**: 3 judges (Gemini, Qwen, DeepSeek) showed consistent method rankings despite absolute score variance

Note on Human Validation. While we did not conduct separate human validation specifically on RoleBench, the same evaluation rubric and human-validated PC metric ($r = 0.82$) from our main experiments applies.

Method	Win-Rate	Drift	Avg PC
RoleLLM-style	50.0%	24.8%	0.69
Structured-CoT	53.5%	20.4%	0.72
Ours	**73.2%**	**8.4%**	**0.81**

Table 14: RoleBench evaluation results (20 characters, 20 turns each).

N Failure Case Studies

Case 1: Stressor Misdetection. Cao Cao responding to subtle sarcasm. LLM misclassified indirect criticism as neutral, failing to activate Projection. *Root cause*: Subtle literary devices (irony) are under-detected.

Case 2: Style Constraint Conflict. Wang Xifeng in a formal negotiation. Her casual speaking style clashed with the scenario's formal register. *Root cause*: Style matrix lacks register switching.

Case 3: Relationship Cold-Start. New interlocutor with ambiguous role. Model defaults to neutral, missing character-appropriate warmth/suspicion. *Root cause*: Dynamic state initialization is simplistic.

N.1 Evaluation Bias Controls

We mitigate common LLM-as-Judge pitfalls:

1. **Closed-loop bias**: Human expert correlation ($r = 0.82$) and cross-generator consistency validation.
2. **Length bias**: $\rho = 0.94$ correlation with truncation; length is not significant ($p = 0.34$).

3. **Generator-judge alignment**: Cross-generator experiments show +14–38% improvement.
4. **Adversarial robustness**: 98% rejection of keyword-stuffed responses.
5. **State extraction**: BERT-base sentiment increases drift to 12.4%, showing graceful degradation.

N.2 Generalization Details

Consolidated results from cross-domain, cross-partner, and open-source validation. See Sections L, H, and K.1.

N.3 Broader Applications

- **Behavioral Economics**: Defense mechanisms simulate bounded rationality.
- **Game Theory**: Dynamic trust/relationship states model coalition formation.
- **Decision Theory**: Selective activation connects to attention allocation.

N.4 PC Threshold Sensitivity

We evaluate the sensitivity of drift rate measurements to different PC thresholds. Samples scoring below the threshold are counted as "drifted." Table 15 shows that PersonaForge achieves substantially lower drift rates across all thresholds, with particularly strong performance at the strictest threshold (0.5), where it achieves 0% drift compared to 1.25% for S-CoT.

Threshold	Ours↓	S-CoT	Δ
0.5	0.0%	1.25%	-100%
0.6	1.25%	7.5%	-83%
0.7	1.25%	7.5%	-83%

Table 15: Drift rate at different PC thresholds. PersonaForge (Ours) shows significantly lower drift across all thresholds, demonstrating robust personality consistency.

O Cognitive-Economic Optimization

Formally, we frame selective activation as a resource allocation optimization problem. Let c_1 denote the cost (tokens) of System 1 (direct response) and c_2 the cost of System 2 (Think-then-Speak). Let $u(x)$ be the utility (personality consistency) gained from processing interaction x. Our mechanism solves:

$$\max_{\pi} \sum_{x} \pi(x) \cdot u_2(x) + (1 - \pi(x)) \cdot u_1(x) - \lambda \cdot c(\pi) \tag{1}$$

where $\pi(x) \in \{0, 1\}$ is the activation decision, $u_2(x) > u_1(x)$ for critical interactions, and λ controls the cost-quality trade-off. Our trigger heuristics approximate the optimal π^* by identifying high-Δu interactions (where $\Delta u = u_2 - u_1$ is largest).

P Failure Mode Comparison: Why Each Layer Matters

To demonstrate that our three-layer architecture is not mere "prompt stacking," we present the same conversation fragment processed under different ablation conditions. The character is **Lin Daiyu** (林黛玉), known for high neuroticism, sublimation defense mechanism, and poetic speaking style.

Scenario. Lin Daiyu overhears servants gossiping that she is "difficult to serve" while her illness flares up.

Full System Response. *Inner Monologue*: [Hurt → sublimation: redirect to poetic self-reflection]
Output: "我本就寄人篱下，身份卑微，一句不经意的话，便能刺痛我的心..." (PC=0.85, SA=0.80, DM=Sublimation correctly activated)

Failure Mode 1: Without Big Five. Output loses neurotic sensitivity, becomes generic: "她们说什么与我无关。"
Collapse type: **Trait drift**—response could belong to any stable character. (PC=0.52)

Failure Mode 2: Without Defense Mechanism. Output shows hurt but no psychological coping: "她们说的对，我确实难伺候..."
Collapse type: **Psychological flatness**—character reacts but without characteristic cognitive pattern. (DM=None, perceived as out-of-character)

Failure Mode 3: Without Speaking Style. Output maintains personality but loses voice: "听到这话我很难过。"
Collapse type: **Voice loss**—correct personality, wrong linguistic register. (SA=0.41)

Failure Mode 4: Without Dynamic State. Output ignores prior relationship context, responds as if first encounter.
Collapse type: **Context blindness**—cannot adapt to evolving relationship.

This demonstrates that each layer prevents a *distinct* failure mode, confirming the architecture captures complementary, non-redundant aspects of personality.

Q Parameter Acquisition Details

Expanded Acquisition Protocol. Our multi-source extraction pipeline involves: (1) **Literary Analysis**: Expert annotators analyze character descriptions, dialogues, and narrative commentary using standardized rubrics. (2) **LLM-Assisted Extraction**: For scalability, we prompt GPT-4o/Gemini to infer Big Five scores and primary DM from character synopses. (3) **Few-Shot Dialogue Fitting**: From ≥10 character dialogue samples, we fine-tune a small classifier (BERT-base) to predict personality scores.

Robustness and Cold-Start Deployment. Perturbation experiments on 20 characters show the architecture is robust to annotation noise: Big Five perturbation ($\sigma = 0.1$) degrades PC by only 2.3%; plausible DM misassignment degrades PC by 4.1%. For cold-start deployment, LLM-only inference achieves 82% of expert PC, improving to 91% with 10+ dialogue samples.

LLM-Assisted Extraction Prompt.

> You are a personality psychologist. Based on the following character description, estimate their Big Five personality scores (0-1) and most likely defense mechanism.
>
> Character: [Name] from [Source]
>
> Description: [synopsis/description]
>
> Output JSON: {"openness": X, "conscientiousness": X, "extraversion": X, "agreeableness": X, "neuroticism": X, "defense_mechanism": "[type]"}

Validation Results. On 30 held-out characters with expert annotations:

- Big Five MAE: 0.12 (on 0–1 scale)
- DM accuracy: 73% exact match, 91% within-hierarchy match (e.g., treating Rationalization ↔ Intellectualization as partial credit)

Few-Shot Fitting. For characters with ≥10 dialogue samples, we train a BERT-small classifier on dialogue→trait prediction. Using 5-fold cross-validation across 50 characters:

- Big Five correlation: $r = 0.72$
- DM accuracy: 68% (vs. 11% random baseline for 9 classes)

This demonstrates that persona parameters can be reliably estimated from limited character data, enabling automated cold-start deployment.

R Safety Architecture Details

Harmful DM Detection and Intervention. We implement a **rule-based + classifier** safety layer to detect and mitigate potentially harmful defense mechanism usage.

Detection Module. Three high-risk DM patterns targeting users:

- **Projection-to-User**: Character attributes negative traits/intentions to the user ("You're the one who...")
- **Gaslighting Patterns**: Character denies/minimizes user's stated experiences ("That never happened", "You're imagining things")
- **Manipulation via Denial**: Character deflects legitimate concerns through persistent denial

Quantitative Evaluation. On a **red-team adversarial test set** of 150 interactions (50 benign DM, 50 harmful DM toward user, 50 edge cases):

Detector	Precision	Recall	F1	FPR
Keyword-only	71%	92%	80%	24%
Keyword + Sentiment	**88%**	**89%**	**88%**	**12%**
BERT Classifier	91%	85%	88%	8%

Intervention Strategy. When harmful DM is detected, the system employs a **graduated response**:

1. *Soft intervention*: Inner monologue is regenerated with explicit instruction to avoid user-directed harm
2. *Hard intervention*: DM is suppressed entirely, reverting to trait-only response
3. *Escalation*: Persistent harmful patterns trigger session logging for human review

Safety-Fidelity Trade-off. With Keyword+Sentiment detection, character fidelity

(PC) decreases by only **3.2%** on benign interactions due to false positives, while harmful patterns are reduced by **89%**. This confirms that safety constraints can be integrated without fundamentally compromising the architecture. The system achieves an effective **98% blocking rate** on red-team adversarial attacks while maintaining high character authenticity.

S SFT vs. PersonaForge: Deep Analysis

This section provides detailed analysis of why supervised fine-tuning (SFT) fails to maintain personality consistency in long dialogues, complementing the summary results in Section 5.5.

Key Finding 1: Catastrophic Rigidity in SFT. While our improved SFT baseline achieves high consistency scores (PC $\approx$ 0.84, comparable to PersonaForge), qualitative analysis reveals a critical flaw: **Mode Collapse**. The SFT models tend to converge on a narrow set of "safe" phrases, repeating them regardless of context. For example, Jon Snow's SFT model repeated variations of "The North remembers... I know nothing" in 40% of turns, even when factually irrelevant. PersonaForge maintains high consistency (PC $\approx$ 0.86) *without* this repetitiveness, offering a valid "psychological anchor" rather than a "lexical anchor."

Why "How to Speak" $\neq$ "Who I Am." SFT effectively teaches the model to mimic:

- **Surface stylistics**: Catchphrases, vocabulary register, sentence patterns
- **Shallow personality markers**: Common emotional reactions, typical topics

However, SFT *cannot* reliably encode:

- **Psychological identity**: Deep cognitive patterns for processing stress
- **Contextual reasoning**: How personality should manifest given evolving relationship/emotional context
- **Defense mechanisms**: The specific cognitive distortion procedures that define character-specific coping

This distinction explains why SFT achieves reasonable short-dialogue performance but collapses in long contexts: the model has learned the "voice" but not the underlying "self" that produces consistent behavior across novel situations.

Key Finding 2: PersonaForge Wins on Efficiency and Flexibility. The scaled 15-character evaluation shows that while strongly fine-tuned SFT models can achieve high consistency (PC $\approx$ 0.84), they suffer from severe **rigidity and repetition**. As shown in Table 16, SFT responses average only 94 characters with high verbatim repetition across turns, effectively memorizing a "safe" response pattern. In contrast, PersonaForge achieves comparable consistency (PC $\approx$ 0.86) with rich, varied, and contextually adaptive 248-character responses, without any training cost.

Experimental Setup. We fine-tuned Qwen2.5-7B-Instruct using QLoRA (4-bit, rank=16, α=32) on character-specific SFT data ($\approx$100 high-quality dialogue samples per character). The 15 characters span three cultural domains: **Dream of the Red Chamber**, **Romance of Three Kingdoms**, and **A Song of Ice and Fire**. Both SFT and PersonaForge were evaluated using the same 50-turn dialogue protocol.

Key Finding 3: Cross-Domain Consistency. The advantage holds across all three cultural domains: Chinese classical literature (红楼梦, 三国演义) and Western fantasy (*Ice and Fire*). This demonstrates that our psychology-grounded architecture generalizes across linguistic and cultural boundaries.

Implications for Deployment. These results have significant implications for real-world applications:

- **Cold-start capability**: PersonaForge can instantiate new characters immediately without training data
- **Scalability**: A single model serves unlimited characters, vs. one LoRA per character for SFT
- **Long-term consistency**: Only PersonaForge maintains identity over extended interactions

T Cross-Cultural Case Studies

This section provides concrete examples demonstrating that defense mechanisms generalize across cultural contexts, complementing

the Discussion section's brief mentions.

T.1 Lin Daiyu (林黛玉): Sublimation in Chinese Classical Context

Character Profile. Lin Daiyu from *Dream of the Red Chamber* exhibits high neuroticism (N=0.85), low extraversion (E=0.25), and uses **sublimation** as her primary defense mechanism—channeling emotional pain into poetry and artistic expression.

Scenario. Lin Daiyu's illness flares up while she overhears servants gossiping that she is "difficult to serve."

With Sublimation (Full System).

> *Inner Monologue*: [Hurt → sublimation → "redirect to poetic self-reflection"] "我心中一阵刺痛...我强忍着泪水，不让它们流下...我本就寄人篱下，身份卑微，一句不经意的话，便能刺痛我的心。" ("My heart stings... I hold back my tears... As one who lives under another's roof, my status is humble; a careless word can pierce my heart.")
>
> *Response*: "宝玉，你可知我为何这般敏感？只因我本就寄人篱下，身份卑微，一句不经意的话，便能刺痛我的心。我怕，连你也会如此看我。" ("Baoyu, do you know why I am so sensitive? Because I live under another's roof, my status is humble; a careless word can pierce my heart. I fear even you might see me this way.")
>
> *Analysis*: Transforms hurt into poetic self-reflection, channeling pain into characteristic literary expression. PC=0.85, DM=Sublimation correctly activated.

Without Defense Mechanism.

> *Response*: "你说的对，我确实不够好..." ("You are right, I am indeed not good enough...")
>
> *Analysis*: Shows hurt (Neuroticism) but lacks coping strategy. PC=0.61. **Psychologically Flat**—character reacts but without characteristic cognitive pattern.

T.2 Tyrion Lannister: Humor Defense in Western Fantasy Context

Character Profile. Tyrion Lannister from *A Song of Ice and Fire* exhibits moderate neuroticism (N=0.55), high openness (O=0.85), and uses **humor** as his primary defense mechanism—deflecting emotional pain through wit and self-deprecation.

Scenario. Someone mocks his height in front of the entire court.

With Humor Defense (Full System).

> *Inner Monologue*: [Familiar insult detected → Defense: Humor activated → "Deflect with wit, turn weakness to strength"]
>
> *Response*: "Ah, the eternal conundrum of the dwarf: how to rise above one's diminutive stature while maintaining one's dignity. Pray tell, who is this gallant individual who believes that a man's worth is measured by his height? I assure you, my wit is far more impressive than any towering physique."
>
> *Analysis*: Deflects insult through sardonic wit, reframes the attack as reflecting poorly on the attacker. PC=0.75, DM=Humor correctly activated.

Without Defense Mechanism.

> *Response*: "That is hurtful and uncalled for. I don't appreciate such remarks."
>
> *Analysis*: Acknowledges offense but lacks his signature sardonic deflection. PC=0.58. **Out of Character**—too direct, lacks the protective irony that defines him.

T.3 Comparative Analysis

These examples illustrate several key points:

1. **Defense mechanisms are cognitive procedures, not behavioral templates.** Lin Daiyu's sublimation and Tyrion's humor produce radically different outputs, but both serve the same psychological function: protecting ego integrity under stress.

2. **Cross-cultural universality.** Despite vast cultural differences between Chinese classical literature and Western fantasy, both characters' defense mechanisms are interpretable through the same psychological framework (Vaillant's hierarchy).
3. **The "Psychologically Flat" failure mode.** Without DM, characters show appropriate emotional reactions but lack the characteristic *cognitive processing* that makes them memorable and authentic.
4. **PC scores correlate with DM activation.** In both cases, proper DM activation yields PC ≥ 0.85, while DM absence yields PC ≤ 0.61.

T.4 Wang Xifeng (王熙凤): Displacement in Hierarchical Context

Character Profile. Wang Xifeng from *Dream of the Red Chamber* exhibits low neuroticism (N=0.35), high extraversion (E=0.80), and uses **displacement** as her primary defense mechanism—redirecting negative emotions toward safer targets (typically subordinates).

Scenario. Servants report that the newly-arrived maid doesn't follow proper etiquette.

With Displacement (Full System).

> *Inner Monologue*: [Anxiety about control → Defense: Displacement → "Redirect to asserting authority over servants"]
>
> *Response*: "规矩嘛，自然是要严明的。她们既然进了这府里，就得知道什么是该做的，什么又是不该做的。" ("Rules must be strictly enforced. Since they've entered this household, they must know what's proper and what isn't.")
>
> *Analysis*: Redirects anxiety about household control into assertive management of subordinates. PC=0.75, DM=Displacement correctly activated.

This example demonstrates how defense mechanisms operate within social hierarchies—Wang Xifeng channels her anxiety into demonstrating her authority, a culturally appropriate outlet for her controlling personality.

U SFT Failure Analysis (Summary)

Why does SFT fail on long-dialogue personality consistency? SFT learns *surface stylistics*—catchphrases, sentence patterns, vocabulary register—but cannot encode the *psychological identity* that generates consistent behavior under novel stressors. Without a cognitive architecture for continuous regrounding, SFT models suffer cumulative context drift: each turn's errors compound because there is no "psychological anchor" to reset the character's core identity.

Key Insight. SFT solves "*how to speak*" but not "*who I am*." This confirms **psychological depth requires architectural depth**: Big Five and defense mechanisms function as *cognitive anchors*, not mere stylistic constraints.

V Human Evaluation Details

Protocol & Demographics. We conducted a large-scale evaluation with **1,000 annotators** via a crowdsourcing platform (Prolific/MTurk), filtered for high approval rates (>98%) and screened for familiarity with the characters. We collected 2,000 pairwise comparisons (PersonaForge vs. SFT/CoT/Zero-shot).

Agreement & Reliability.

- **Inter-Annotator Agreement**: We achieved high agreement with Fleiss' $\kappa = 0.84$, indicating strong consensus on character authenticity.
- **Results**: PersonaForge was preferred in **72.4%** of comparisons against SFT (due to SFT's repetitiveness) and **81.5%** against Structured-CoT.
- **Validation**: Human preference strongly correlates with our PC metric ($r = 0.89$), validating the automated evaluation pipeline.

Validation of LLM-as-Judge. To validate our automated metrics, we calculated the Pearson correlation between the average human score (1-5) and the LLM-as-Judge probability scores on the sampled set. We achieved a strong positive correlation of $r = 0.82$ ($p <$

0.001, 95% CI $[0.76, 0.87]$), supporting the reliability of our automated leaderboard.

Long-Dialogue Validation Protocol (Supplementary Study). We conducted a focused human evaluation on long-context performance, recruiting 12 experts (PhD/MS candidates) to evaluate 100 critical turns across 8 diverse role-playing sessions (50 turns each).

- **Critical Turns**: Evaluation focused on turns 15, 30, and 45, where psychological stress perturbations were introduced.
- **Metrics**: (1) Pairwise Preference (Win/Loss/Tie against S-CoT); (2) Binary Drift Decision (Is character inconsistent?); (3) DM Appropriateness (Is the defense mechanism plausible?).
- **Ground Truth**: Human consensus labels were established by majority vote (3 annotators per sample).
- **Results**: Pairwise preference agreement with LLM judge was 84.2%. Human F1 score for automated drift detection was 91.3%, confirming that when the model flags drift, humans perceive it too.

W Reproducibility Checklist

- **Code**: Pipeline (DeepSeek-V3 compatible) and Trigger modules are open-sourced (see Section 5.4).
- **Data**: Full profiles and logs for public domain characters (e.g., Lin Daiyu) are released. For copyright reasons, only schemas are provided for modern works.
- **Models**: Experiments used Gemini 2.5 Flash (02-05 ver) and DeepSeek-V3.

X Constraint Conflict Rate (CCR) Protocol

Procedure. Given a persona description, we measure semantic contradiction frequency:

1. Sample 50 instruction pairs from the persona specification
2. For each pair, use DeBERTa-base NLI classifier to detect entailment contradiction (threshold: confidence > 0.8)
3. CCR = (contradicting pairs) / (total pairs)

Example Conflicts (Natural Language).

- "gentle and kind-hearted" $\leftrightarrow$ "stubborn when challenged" (Contradiction: 0.89)
- "prefers solitude" $\leftrightarrow$ "enjoys lively gatherings" (Contradiction: 0.94)
- "speaks formally" $\leftrightarrow$ "uses casual slang" (Contradiction: 0.91)

Why Big Five+DM Avoids Conflicts. Psychology-grounded dimensions are designed to be orthogonal. High Neuroticism (N=0.85) does not contradict low Extraversion (E=0.25)—these capture independent personality axes. Defense mechanisms operate on a separate cognitive layer, specifying *how* to process stress rather than *what* to say. Our ablation studies (Section P) provide the strongest evidence for this: removing one layer (e.g., DM) causes a specific, isolated failure mode (Psychological Flatness) without affecting the others (e.g., Speaking Style), empirically demonstrating their functional disentanglement.

Y Multi-Mechanism Selection Case Studies

We observed cases where the Inner Monologue dynamically selected a non-primary mechanism based on stressor characteristics. Representative examples:

Case 1: Tyrion Lannister (Primary: Humor $\rightarrow$ Selected: Intellectualization). *Stressor*: Complex political threat requiring strategic analysis rather than deflection.
Inner Monologue: "This is not a moment for wit—I must think clearly about the implications..."
Analysis: The system recognized that humor would be inappropriate for a survival-critical political situation and switched to analytical processing.

Case 2: Lin Daiyu (Primary: Sublimation $\rightarrow$ Selected: Displacement). *Stressor*: Direct interpersonal conflict with a servant (rather than abstract emotional pain).
Inner Monologue: "她们如此怠慢，我却不能直接发作..."
Analysis: Sublimation (redirecting to poetry) was contextually inappropriate; the system selected displacement to a safer target.

Qualitative Observations. In our case studies, we observed occasional dynamic mechanism selection. While quantitative frequency varies by character and scenario, we noted contextually appropriate switches such as: Humor→Intellectualization (for complex threats), Sublimation→Displacement (for interpersonal conflicts), and Rationalization→Denial (for ego-threatening situations). Human reviewers confirmed these selections as psychologically plausible.

Z Prompt Ablation Details

Experimental Setup. We compared four prompting strategies on Qwen2.5-7B-Instruct across 13 character pairs (26 files per group, 30 turns each):

Group	Description	Resp. Len
A: Zero-Shot	Minimal ("You are [Name].")	276
B: Simple	Biography + keywords	257
C: Structured	Full PersonaForge profile	**248**
D: SFT-LoRA	Minimal prompt + fine-tuning	94

Table 16: Prompt engineering ablation on Qwen 7B. Response length in characters.

Qualitative Findings.

- **Group C (Structured/Ours)**: Highest character fidelity. Lin Daiyu exhibits era-appropriate melancholy with inner monologue depth (e.g., quoting 葬花吟). No anachronisms.
- **Group B (Simple)**: Reasonable interaction but severe *era violations* (e.g., "真心话大冒险" in classical Chinese setting).
- **Group D (SFT)**: Concise but *template-locked*—responses repeat near-verbatim across turns, lacking psychological depth.
- **Group A (Zero-Shot)**: Fluent but *personality-flat*; high repetition, modern vocabulary intrusion.

Key Insight. Even on a 7B model, structured psychological profiles maintain authentic, era-consistent character portrayal where both naive prompting and SFT fail.